

Daanesh Potnuri

📞 717-686-6770 ✉ daaneshpotnuri.3690@gmail.com 🔗 [linkedin.com/in/daanesh-potnuri](https://www.linkedin.com/in/daanesh-potnuri) 🌐 github.com/Daanesh

Summary

Full-stack engineer skilled in modern frontends, secure backends, and API-driven microservice architectures, with AWS/GCP delivery experience. Integrates AI/LLM pipelines to build fast, reliable, and intelligent user experiences.

Education

The Pennsylvania State University

08/2023 - 05/2025

Masters of Science in Computer Science, **CGPA: 3.36/4.0**

Pennsylvania, USA

Experience

Software Developer – World Disaster Center

07/2025 – Present

- Delivered a searchable, filterable roster with server-side pagination and CSV export using React, Node.js/Express, and PostgreSQL; reduced data load time by 35% and cut admin update time by 30 %.
- Built a lightweight CMS with role-based access and draft→publish flow (JWT-secured REST APIs); increased publishing throughput by 40% and reduced turnaround from 2 days → same day.
- Migrated legacy endpoints to clean REST services, normalized schemas, and added indexing/connection pooling; improved median response time by 25% and lowered API error rate by 60% (uptime 99.9%).
- Integrated GA4 and custom event logging; built a simple KPI dashboard (page views, CTR, active users) that informed UX fixes leading to +20% session duration and +15% click-through to solutions pages.

Graduate Research Assistant & Software Developer – The Pennsylvania State University 08/2023 – 05/2025

- Architected a Python full-stack data labeling app (FastAPI, PostgreSQL/SQLAlchemy, JWT, Docker), automated CSV ingest and uploads, and orchestrated an admin dashboard—streamlined prep time by 50%.
- Spearheaded a React + Node/Express portal with server-side pagination, advanced filters, and role-based views; optimized workflows to reduce TA overhead by 30% and improve student turnaround.
- Designed and operationalized a Transformer + BiLSTM forecasting pipeline (NYC) in TensorFlow/Keras; benchmarked results (RMSE 34.1, MAE 23.7, MAPE 4.1%, R^2 0.92) and automated ETL, evaluation, and CI.

Associate Software Engineer – Mphasis

08/2020 – 08/2023

- Built & optimized Java REST services for the Fircosoft sanctions platform using Spring Boot + Hibernate with Oracle/PostgreSQL, improving data ingestion and reducing API latency by 25%.
- Standardized delivery by containerizing 12 microservices with Docker/Kubernetes and automating CI/CD via Jenkins/Git, cutting deployment failures by 40% and environment drift by 50%.
- Hardened reliability with Splunk + Prometheus/Grafana observability and Tomcat release hygiene, reducing incident response time by 35% and sustaining 99.9% uptime.
- Authored Unix shell scripts to automate routine QA tasks, elevating team productivity by 40%.
- Accelerated QA by automating 200+ test cases (JUnit/Selenium) and leading performance tuning, lowering system latency by 35% and improving compliance accuracy by 20%.

Skills Summary

Languages: Java, Python, TypeScript/JavaScript, SQL

Frontend: React, Next.js, HTML/CSS, Tailwind; SPA state server-side pagination; form validation

Backend & APIs: Spring Boot (REST, JPA/Hibernate), Node.js/Express, FastAPI; API design, JWT/RBAC, caching

Databases: PostgreSQL, Oracle, MySQL, MongoDB, Redis; schema design, indexing, query optimization; ORMs (Hibernate, SQLAlchemy)

Cloud DevOps: Docker, Kubernetes, Jenkins, GitHub Actions, AWS (EC2, S3, RDS, IAM), Vercel; CI/CD, env promotion, IaC (Terraform – basics)

Quality, Observability Security: JUnit, Selenium, PyTest, Postman; performance/load testing, coverage, regression; Splunk, Prometheus/Grafana, Sentry, GA4; OAuth2/JWT, RBAC, secrets mgmt, rate limiting, secure APIs; Git, Agile/Scrum, PR reviews, Swagger/OpenAPI, Jira

Projects

RAG-Based QA Bot — Musical Instruments E-commerce | *PyTorch, DistilBERT, FAISS, FastAPI, LangChain, Ollama, Ng*

- * Engineered a domain-specific RAG chatbot over the Amazon Reviews 2023 dataset (213K+ products) using DistilBERT, FAISS, and LangChain for retrieval and response generation. Integrated sentiment analysis and optimized retrieval to deliver accurate, real-time product insights with 2–3 s latency and 95% off-topic query filtering.

Decentralized File Sharing dApp | *Blockchain, Solidity, React.js, IPFS*

- * Architected a secure peer-to-peer file-sharing dApp on Ethereum, implemented Solidity contracts for access control and metadata, and integrated IPFS (Pinata) with a React + Web3.js frontend—accelerating onboarding by 80%, maintaining 99.99% uptime, and validating smart-contract security with 95% test coverage.

Automated Examination System | *Python, Flask, NLP, Machine Learning*

- * Operationalized an NLP pipeline to auto-generate and grade multiple-choice questions, integrated into a Flask web app for instructor review/edit/export—IJITEE (2020) publication—cutting manual authoring time by 30%.